

A Reusable Edge AI System for Multimodal AI Companion: Performance and Response Evaluation

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Abstract—With the advancement of edge computing and artificial intelligence, multimodal companion systems have demonstrated significant potential in mental health assistance. However, executing continuous visual perception and large language models (LLMs) simultaneously on resource-constrained edge devices poses severe computational and memory challenges. This study develops a dynamically decoupled Multimodal AI Companion system based on the Kneron Kneo Pi. Through heterogeneous hardware allocation and an asynchronous state machine, the system successfully integrates visual emotion recognition, audio output, and dynamic prompt engineering on a single edge device. The experiments quantitatively analyze the multidimensional trade-offs among accuracy, latency, memory footprint, and dialogue quality for “lightweight vs. modern visual models” and “local vs. offloaded LLM deployments,” providing practical design guidelines for future deployment of edge AI companion systems.

Keywords—AI Companion, Edge AI, Large Language Models (LLMs), Multimodal Companion System, Vision Perception.

I. INTRODUCTION

Multimodal AI companion systems play a crucial role in human-computer interaction (HCI). Traditional systems rely heavily on cloud computing to process images and speech. This not only raises severe privacy leakage concerns but also disrupts the immersion of interactions due to unpredictable network latency. Therefore, deploying AI models on edge devices has become an inevitable trend.

However, on hardware-constrained edge devices, the concurrent operation of modern high-precision visual models and LLMs is highly prone to out-of-memory errors or severe system latency. Existing research primarily focuses on compressing single models, lacking exploration of the overall resource scheduling of multimodal systems. Particularly in continuous companion scenarios where real-time emotional perception and empathetic dialogue are strictly required, a holistic resource management framework is urgently needed. To address this challenge, this study proposes a reusable multimodal edge AI system.

II. RELATED WORK

Recent studies [1]-[5] have shown that AI companions are evolving from task-oriented conversational agents into socially aware systems intended to support long-term affective and interpersonal engagement. Chaturvedi *et al.* [1] reviewed the literature on social companionship with artificial intelligence and argued that emotional bonding, social presence, anthropomorphism, and trust are becoming core design dimensions of companion systems. From a well-being perspective, Nieves *et al.* [2] further emphasized that digital companions should not be evaluated solely by interaction fluency, but also by personalization, empathetic support, activity and conversation tracking, portability, compatibility, and ethical accountability. In parallel, emotional support

conversation has become an important research direction. Liu *et al.* [3] introduced the Emotional Support Conversation task and the ESConv dataset, showing that emotionally supportive dialogue requires not only fluent generation but also explicit support strategies. This line of work was extended by AugESC [4], which improved topic diversity and generalization by augmenting dialogue with large-language models. More recently, broader companion-AI scholarship has also highlighted the relational risks and benefits of these systems, arguing that companion AI should be designed with explicit attention to how it shapes human relationships and social behavior [5]. Collectively, these studies suggest that AI companions should be assessed not only by conventional language quality metrics but also by emotional appropriateness, user trust, perceived companionship quality, and well-being-oriented system responsibility.

A second stream of recent research focuses on multimodal emotional understanding and the generation of emotionally aware responses. Chu *et al.* [6] proposed a multimodal emotional support conversation framework and the MESC dataset, integrating text, audio, and video to jointly model user emotion, support strategy, system emotion, and response generation. Wang *et al.* [7] further improved multimodal emotional support conversation by introducing reinforcement learning to better align system reasoning and response quality with emotional and strategic objectives. On the perception side, Kovačević *et al.* [8] studied multimodal emotion recognition for human-chatbot interaction in realistic settings. They demonstrated the importance of combining visual, acoustic, and textual cues in the wild. Recent multimodal emotion recognition research has also moved toward stronger fusion architectures, including cross-modal transformer-based methods such as MemoCMT [9]. At the survey level, Wu *et al.* [10] showed that multimodal emotion recognition in conversations is increasingly oriented toward robust fusion, contextual modeling, and real-world conversational deployment rather than isolated benchmark performance alone. However, despite this progress, most prior studies still emphasize either emotional perception or supportive response generation as separate modules, while relatively few provide a reusable end-to-end companion architecture that jointly considers multimodal perception, supportive interaction, well-being-oriented design constraints, and efficient edge deployment. This limitation motivates the proposed reusable edge AI companion system for multimodal emotional companionship. The main contributions of this study are as follows:

- **Dynamic System Architecture Design:** Proposes a resource-decoupled state machine architecture and achieves fluent multimodal edge interactions through CPU/NPU heterogeneous task allocation and HTTP communication.

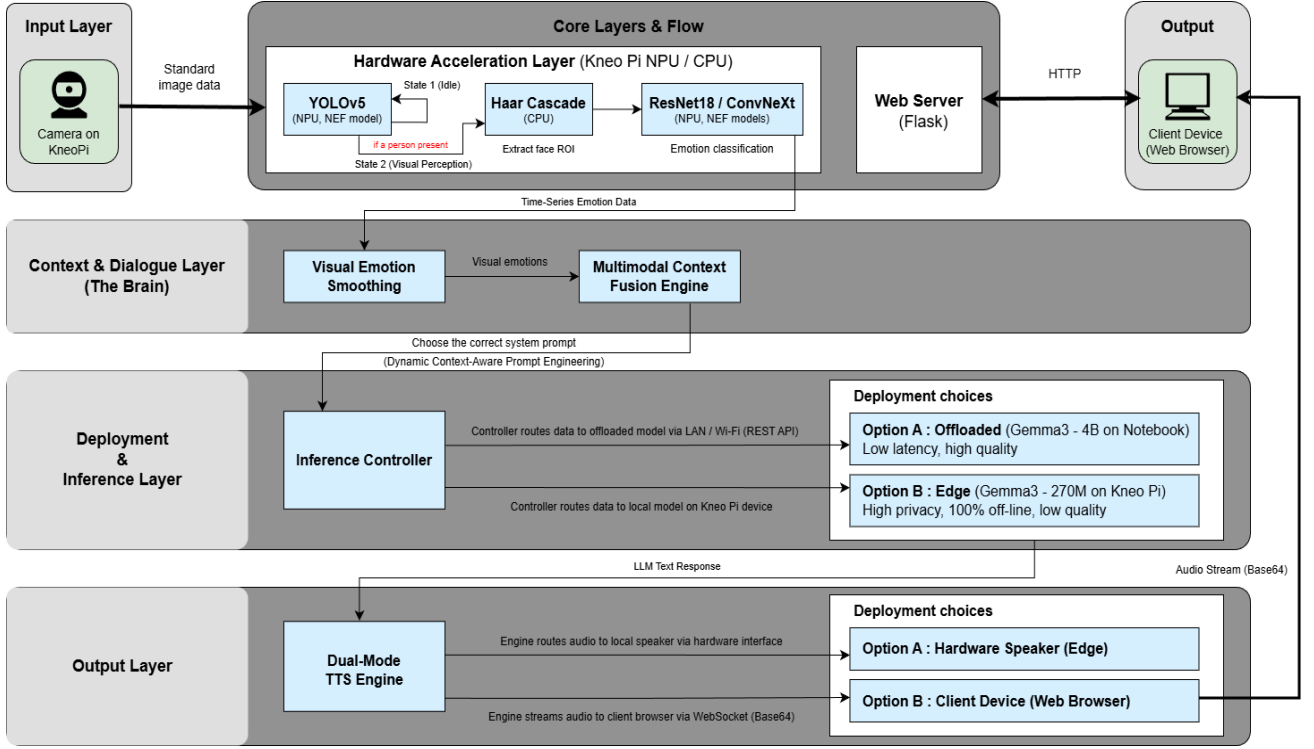


Fig. 1. Overall architecture of the decoupled multimodal AI companion system

- Trade-off Analysis of Visual Perception Modules: Quantitatively compares the hardware throughput and static accuracy of ResNet18 and ConvNeXt on the KL730 NPU.
- Trade-offs in LLM and Output Routing: Provides an in-depth analysis of the differences in system latency, available memory footprint, and audio output flexibility between “full-edge local” and “distributed collaborative” deployments.

III. SYSTEM ARCHITECTURE

The system adopts a CPU/NPU heterogeneous hardware allocation strategy, offloading intensive tasks (YOLOv5, FER) to the Kneron Kneo Pi with KL730 NPU while retaining logic and preprocessing (Haar Cascade) on the CPU. As shown in Fig. 1, the overall architecture comprises four core layers:

A. Dynamic State Machine and Asynchronous Vision Pipeline

To maximize computational efficiency and reduce power consumption, the system introduces a three-stage lifecycle management:

- State 1 (Idle): Upon system initialization, the lightweight YOLOv5 model [11] is prioritized on the NPU for human presence detection. This model was selected due to its highly optimized architecture for constrained edge deployment scenarios [12]. If no user is detected, the system remains in the Idle (Loop) state.
- State 2 (Visual Perception): Once a user is detected, the system wakes the Haar Cascade [13] on the CPU to extract the facial Region of Interest and immediately drives the NPU to execute the emotion recognition model (ResNet18 [14], [15] or ConvNeXt [16]). This

pipeline employs an asynchronous frame-dropping mechanism to prevent the main system program from being blocked by NPU inference latency.

- State 3 (Multimodal Interaction): After the vision pipeline generates the Time-Series Emotion Data, the system enters the decision-making process.

B. Multimodal Context Fusion Engine

Serving as the “brain” of the system, this module bridges visual perception and semantic generation. First, the system filters single-frame noise using the Visual Emotion Smoothing step. Subsequently, based on the smoothed emotion results and historical dialogue records, the fusion engine applies Dynamic Context-Aware Prompt Engineering to automatically replace the system prompt, better aligning it with the current emotional state and ensuring the LLM generates an empathetic response.

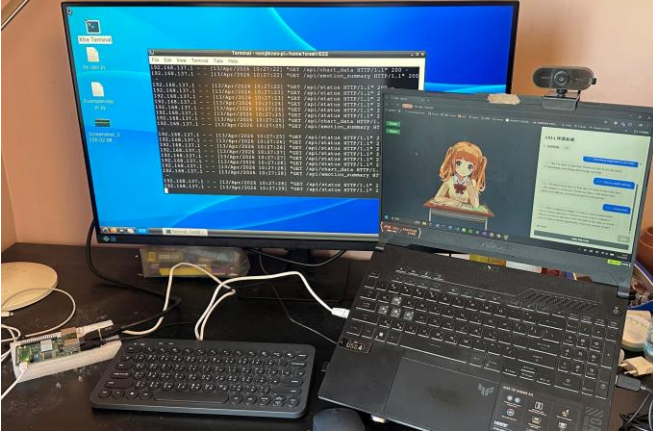
C. Interoperable Dual-Brain Deployment and Routing

The system features a highly flexible Inference Controller, allowing users to switch the deployment architecture based on the resource environment dynamically:

- Option A (Offloaded): Transmits inference tasks via a local area network (LAN/Wi-Fi) to a high-performance node on a laptop (e.g., Gemma3-4B [17]) to achieve ultra-low latency and high dialogue quality.
- Option B (Edge): Executes a lightweight model (e.g., Gemma3-270M [17]) directly on the local Kneo Pi, ensuring privacy and system survivability even in a completely disconnected offline environment.

D. Headless Hub and Dual-Mode Output Layer

Through a Flask web server, the system dispatches the user interface to the Client Device browser and establishes



(a)



(b)

Fig. 2. Hardware deployment: (a) Close-up of the Kneo Pi edge device; (b) Physical system setup during real-time interaction.

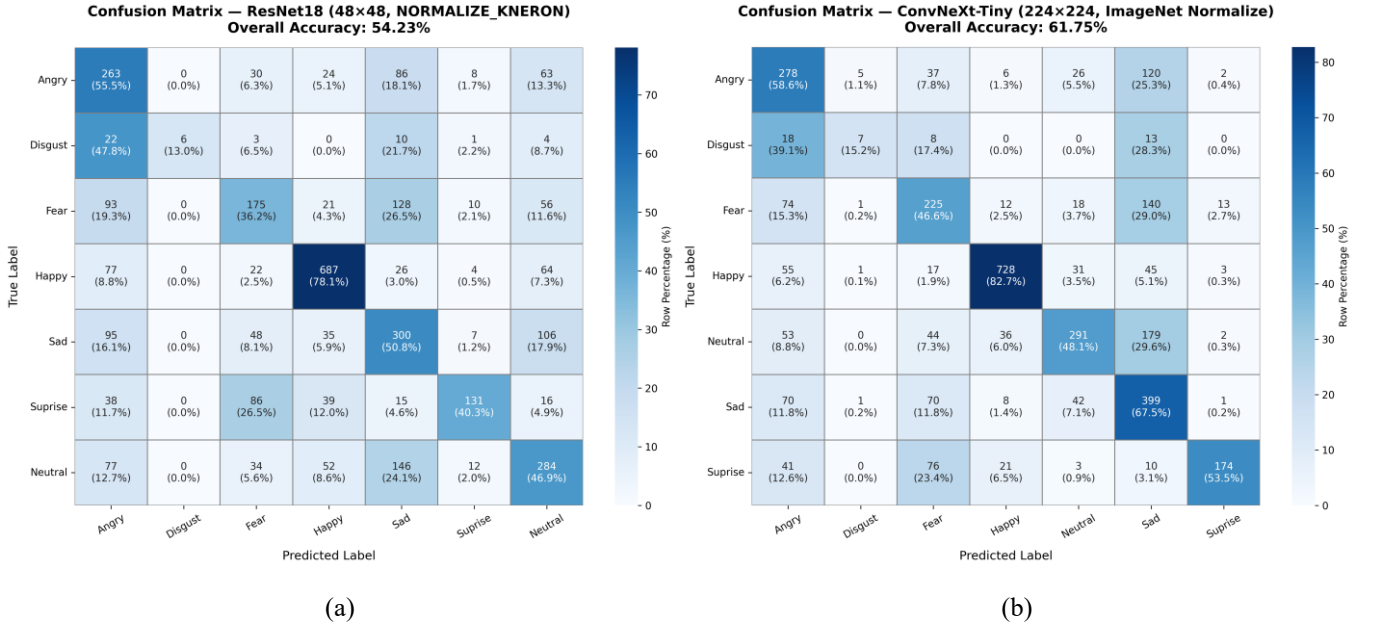


Fig. 3. Confusion matrices on the FER2013 test set: (a) ResNet18 and (b) ConvNeXt-Tiny. The color intensity represents the classification density. ConvNeXt-Tiny shows a more concentrated diagonal, particularly in the "Happy" and "Neutral" categories.

communication via HTTP. For audio output, the system integrates a Dual-Mode TTS Engine. It can dynamically route audio to the Kneo Pi's local audio output. If no external speaker devices are available during deployment, the audio stream can also be transmitted in Base64 format to the client browser for playback, significantly enhancing the resilience of hardware deployment.

The system is implemented on the Kneo Pi edge platform, featuring a compact hardware footprint as shown in Fig. 2(a). The complete prototype deployment, including the visual capture and user interface terminal, is illustrated in Fig. 2(b).

IV. EXPERIMENTAL EVALUATION

A. Experiment 1: Performance and Accuracy Trade-offs of the Visual Perception Module

Both visual models were trained and evaluated using the FER2013 dataset [18]. As summarized in Table I, ConvNeXt-

Tiny achieved a higher overall test accuracy (61.75%) compared to ResNet18 (54.23%). To further investigate the classification performance across seven emotional categories, the confusion matrices for both architectures are presented in Fig. 3.

- Evaluation 1: ConvNeXt achieved an overall accuracy of 61.75%, outperforming ResNet18's 54.23%. To further investigate the classification performance across emotional categories, the confusion matrices are presented in Fig. 3. Both models excel at identifying *Happy* expressions (ConvNeXt F1-score: 0.86 vs. ResNet18: 0.79). However, a distinct "Confusion Zone" exists between *Sad* and *Neutral* states because of subtle similarities in facial microexpressions. ConvNeXt demonstrates superior robustness in distinguishing these negative emotions, whereas ResNet18 exhibits a higher tendency to misclassify *Fearful* inputs as *Surprise*.

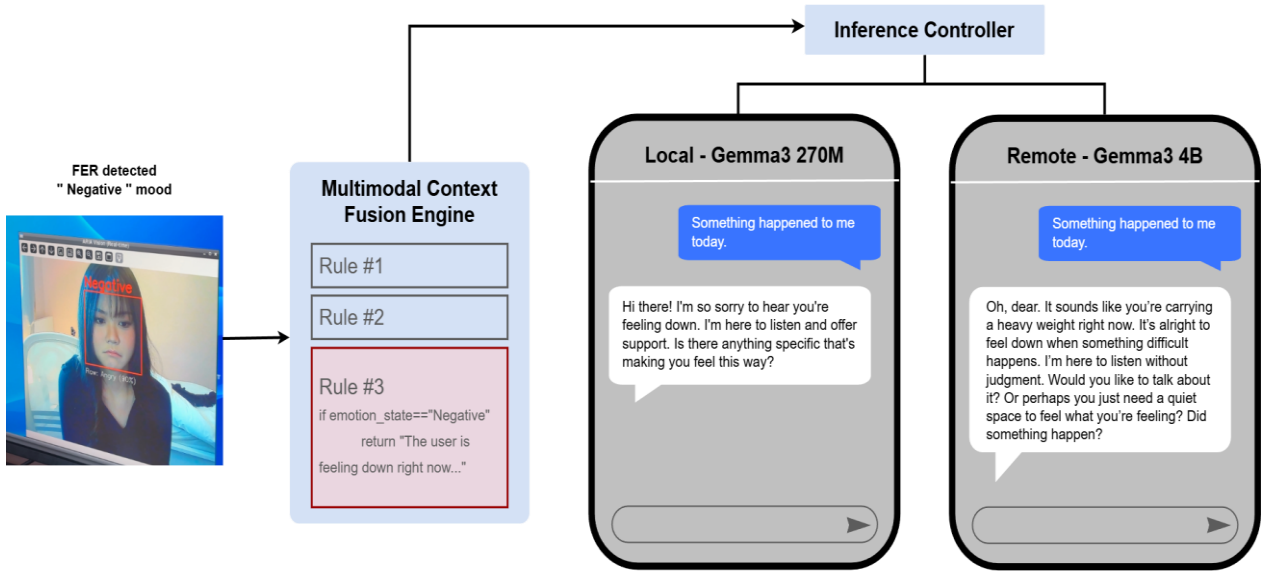


Fig. 4. Qualitative comparison of dialogue generation between Local Edge (270M) and Remote Offloaded (4B) deployments.

TABLE I. HARDWARE PERFORMANCE AND ACCURACY

Metric	ResNet18 (Lightweight)	ConvNeXt (Modern)
Overall Accuracy	54.23%	61.75%
NPU Inference Latency	24.61 ms	54.42 ms
Hardware Throughput	40.65 FPS	18.37 FPS
RAM Footprint	125.28 MB	140.23 MB

TABLE II. PERFORMANCE COMPARISON OF LLM DEPLOYMENT

Architecture	LAN Offloading	Full Edge Local
Model Executed	Gemma3-4B	Gemma3-270M
Average Latency	1.1 ~ 1.5 s	2.5 ~ 4.2 s
RAM Footprint	~0 MB (Offloaded)	~600 MB (mmap cache)
Average Response Quality	4.67	3.6

- Evaluation 2 : Dynamic Real-time Performance: The ResNet18 NPU latency was merely ~24.6ms (40.6 FPS), whereas ConvNeXt recorded ~54.4ms (18.3 FPS). (Note: The drop in System FPS during the experimental video testing was due to the strict sequential frame-by-frame sampling setup. In actual deployment, benefiting from the asynchronous architecture and NPU heterogeneous allocation, the system maintains smooth real-time interactions.)

Consequently, while ConvNeXt provides superior accuracy for fine-grained emotion analysis, its increased inference latency poses a burden on edge hardware. This highlights the necessity of the proposed decoupled architecture, which allows developers to dynamically select between lightweight and modern models based on the real-time constraints of specific application scenarios

B. Experiment 2: Deployment Scenario Trade-offs of the Language Generation Module

- Evaluation 1: System Latency and Resource Burden: As illustrated in Table II, the LAN offloading architecture achieves lower latency and zero additional memory burden, whereas the full edge deployment ensures absolute offline capability at the cost of a 600MB memory footprint and higher latency.
- Evaluation 2: Dialogue Quality and Output Flexibility: A qualitative dialogue example using a chat interface representation is illustrated in Fig. 4. To quantitatively assess the conversational capability, a heuristic evaluation was conducted using a 5-point Likert scale based on two dimensions: *Empathy* and *Coherence*. A score of 1 indicates an irrelevant or unempathetic response, a score of 3 represents basic relevance with minimal comfort, and a score of 5 denotes highly empathetic and coherent long-text generation. Given the prototype nature of the edge system, the authors, serving as expert evaluators, conducted an initial heuristic evaluation. As summarized in Table II, the offloaded 4B model achieved an average score of 4.67, excelling in deep emotional support. Conversely, the edge-based 270M model scored an average of 3.6. Although it generates shorter text, the dynamic prompt compensation ensures it reaches the baseline of acceptable companion quality by providing accurate emotional comfort.

C. Experiment 3: Overall System Integration

This experiment evaluates the sustainability and total responsiveness of the integrated system. While previous sections focused on individual modules, this section examines how the system orchestrates resources across the three operational states defined in Section II-A.

As illustrated in Fig. 5 and Fig. 6, the dynamic state machine effectively manages hardware resources. During State 1 (Idle), the system maintains a low-power footprint with the CPU at approximately 170% (aggregated across the quad-core architecture, where 400% represents full

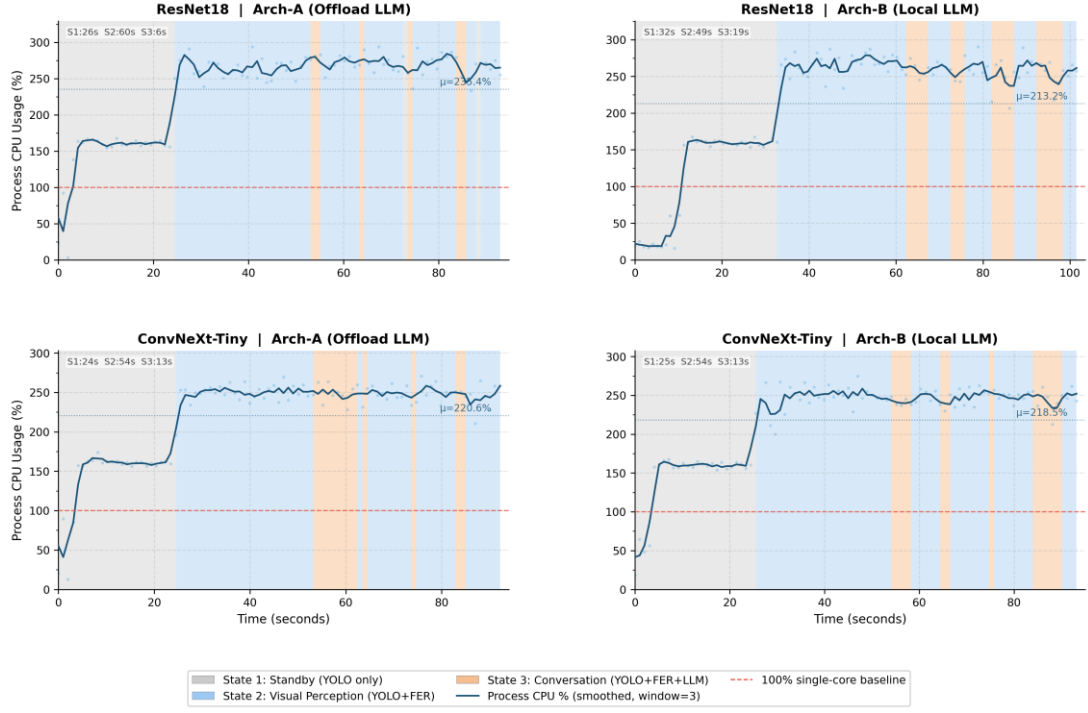


Fig. 5. CPU utilization during system state transitions.

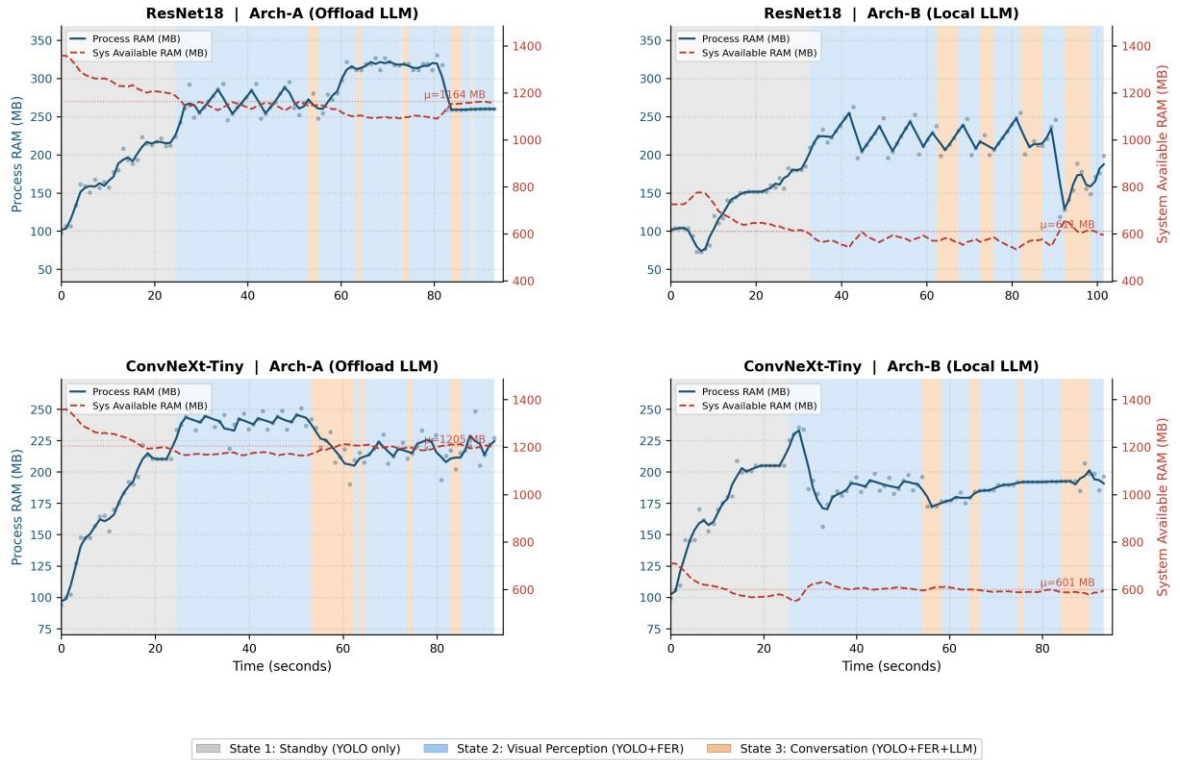


Fig. 6. RAM footprint comparison between different architectures.

utilization) and RAM at ~280 MB. Upon entering State 3, the "Decoupled Architecture" prevents resource exhaustion. Specifically, Fig. 5 shows that while the full-edge deployment (Gemma3-270M) pushes RAM usage close to its physical limit (~1.1 GB total), the offloaded architecture maintains a stable, healthy memory margin, ensuring long-term operational stability.

V. CONCLUSION

This study successfully developed and deployed a Multimodal AI Companion system with dynamic context awareness on a resource-constrained Kneron Kneo Pi edge device. Extensive experiments quantified the practical trade-offs between modern and lightweight visual perception models, as well as the impacts of local versus offloaded LLM deployments.

Furthermore, overall system profiling validated the robustness of the decoupled state machine, demonstrating its ability to sustain hardware memory limits and efficiently manage end-to-end latency. Ultimately, the interoperable architecture and headless hub design of this system provide essential theoretical and practical guidelines for the future deployment of AI companion robots across diverse scenarios, ranging from privacy-critical medical facilities to performance-oriented smart homes.

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